

3.2 The threats to the Antarctic ecosystem and how they can be managed

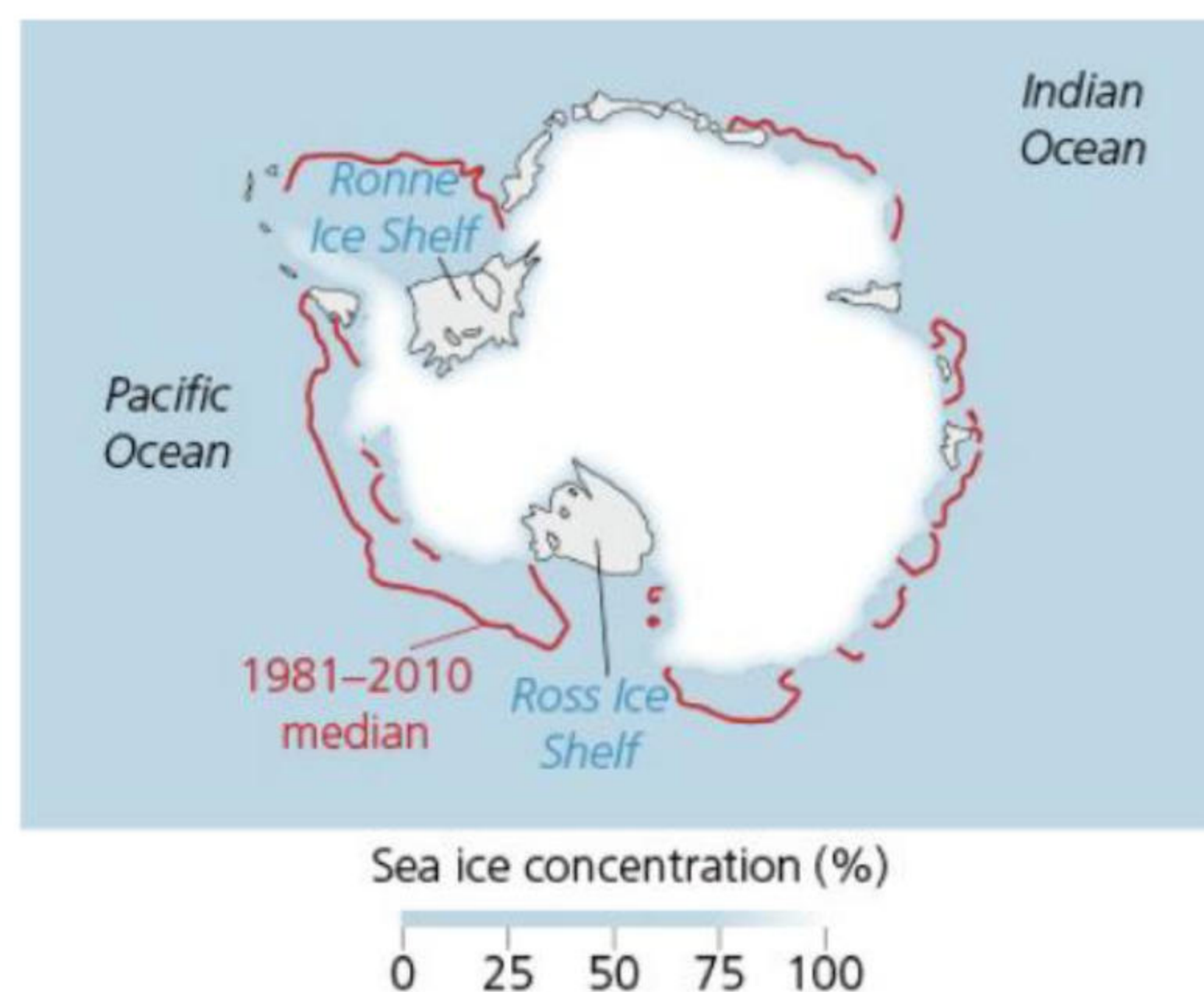
This chapter will explain:

- ★ the threats posed to Antarctica
- ★ the impacts of the destruction of the Antarctic ecosystem
- ★ to what extent the strategies and techniques used to manage the Antarctic ecosystem have been successful.

The threats posed to Antarctica

Climate change

Global climate change and the resultant melting ice sheets are major problems for Antarctica, in particular West Antarctica. In February 2023, Antarctic sea ice extent set a new record low ([Figure 3.6](#)), putting at risk the species that depend on stable sea ice as a breeding habitat. In 2023, Australian scientists reported that the strong ocean currents in the Southern Ocean could slow by more than 40 per cent by 2050. Moreover, the currents have already slowed by about 30 per cent since the 1990s. The reason is the melting taking place at the edge of Antarctica, which is adding more fresh water into the Southern Ocean, reducing its density. This reduces its ability to sink and to drive the deep flows that spread nutrients around the world.



▲ **Figure 3.6** The decline of ice around Antarctica, 1981–2010 compared with 2023

This has occurred with a global warming of about 1.1°C. If warming rises by 2°C – which is predicted by some to occur by 2100 – this would lead to major changes in the ocean and to the continent. It is not just ice at the edges that is melting, the whole ice cap is losing ice. Warmer water below the West Antarctica ice sheet is causing thinning from below.

Global warming is also having an impact on emperor penguins, as they breed and raise their young on sea ice. They are especially vulnerable to the loss of sea ice, to rising temperatures and altered wind systems. It is estimated that a rise in temperature of 1.5°C could lead to a decline of 19–31 per cent in emperor penguin colonies – but if global warming continues (business as usual) to 2100, up to 80 per cent of emperor penguin colonies could become extinct.

Resource exploitation

Mining and mineral exploitation are prohibited on Antarctica. However, scientific investigations into biotic resources are increasing. Some genetic and biochemical resources have been used to develop krill oil, food-related products and anti-freeze proteins. It is estimated that there are over 200 companies and research organisations involved in biological prospecting in Antarctica.

Fishing

Fishing in Antarctic waters is allowed and is increasing as fish stocks in other seas are increasingly overfished. Fishing practices can lead to a decline in predator species, increasing unintentional catch of juvenile fish, destruction of habitat by trawlers, and the death of non-target species caught up in nets and longlines. Longline fishing involves lengthy fishing lines with up to 20,000 baited hooks, which may also capture sea birds as they fish. Overfishing has led to the decline of species such as Antarctic hake and

Patagonian toothfish. Krill is endemic to Antarctic waters and many species depend on it as their main food source. However, krill is sensitive to rising ocean temperatures and it is also impacted by large-scale fishing and pollution in oceans.

Tourism

Antarctica receives upwards of 100,000 tourists per year. Transporting people to Antarctica involves emissions of 'black carbon' and other greenhouse gases by ships and aeroplanes. In addition, there is a risk of non-native insects, seeds and spores being carried in on vessels or by passengers. Most of the tourism pressure is on the Antarctic Peninsula, where 90 per cent of the tourism is concentrated.

Before the Covid-19 pandemic, some 74,000 tourists travelled to Antarctica, and around 20,000 remained on cruise ships visiting the continent. The numbers dropped to 23,000 during the pandemic but since then numbers have increased to around 100,000, with 75,000 'shore' visits.

Pollution

Pollution is an increasing problem. Microplastics are washed in from surrounding islands and seas, or they may be brought in by tourists or scientists. Sea birds often mistake plastic for food and eat it or feed it to their young. The remains of fishing equipment have been washed up on nearby islands. Seals and penguins are known to have become entangled in fishing nets, and the chances of rescue or release are slim.

Activities

- 1 Explain why the emperor penguin is vulnerable to global climate change.
 - 2 Explain why plastics and fishing equipment are a threat to Antarctica's ecosystems.
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The impacts of the destruction of the Antarctic ecosystem

There has been continued human presence throughout the most biodiverse parts of Antarctica over the past 200 years. This has led to infrastructure developments, vegetation trampling, sealing, whaling, fishing, wildlife and soil disturbance, pollution and the introduction of alien species. However, human activity in Antarctica should be easier to regulate than elsewhere because it is restricted to science, television programmes and tourism, with few other human activities or permanent inhabitants.

Climate change is predicted to have major impacts on Antarctic ecosystems. For example:

- » Rapid warming in the west Antarctic Peninsula and a change from snowfall to summer rain has led to an increase in plant communities across the Antarctic Peninsula on newly exposed land, as well as the introduction of 'alien' organisms such as grasses, flies and bacteria.
- » Loss of sea ice west of the Antarctic Peninsula has led to a decline in krill and Adélie penguins. As krill is a major source of food for many marine animals and sea birds, this could lead to the decline of numbers of many species. The decline of sea ice means that species that rely on it are seeing their habitat shrink.
- » Models suggest a 3°C increase in temperature by 2100, which should increase phytoplankton productivity.

Historical changes are also important – for example, the whaling industry led to the near extinction of blue whales and baleen whales in the Southern Ocean. The extinction or near extinction of any single species has a knock-on effect on other species and contributes to loss of biodiversity. This can be seen here, as with less competition for krill from whales, the number of fur seals increased dramatically in just 20 years, from a few hundred to over 20,000. The seals came onshore at Signy Island in the South Orkney Islands, and caused much vegetation destruction through trampling, and eutrophication from the large amount of excrement left on the island.

Activities

- 1 Outline the impacts of global climate change on Antarctica's ecosystems.
 - 2 Explain how changes in the whaling industry led to ecosystem destruction on Signy Island.
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An evaluation of the strategies and techniques used to manage the Antarctic ecosystem

International agreements

The main strategy for managing Antarctica has been the 1959 Antarctic Treaty. This established a system for international cooperation, especially in the name of science, and the demilitarisation of Antarctica. It also prohibits all forms of mining and mineral exploitation. This enables sustainable management of Antarctica's ecosystems.

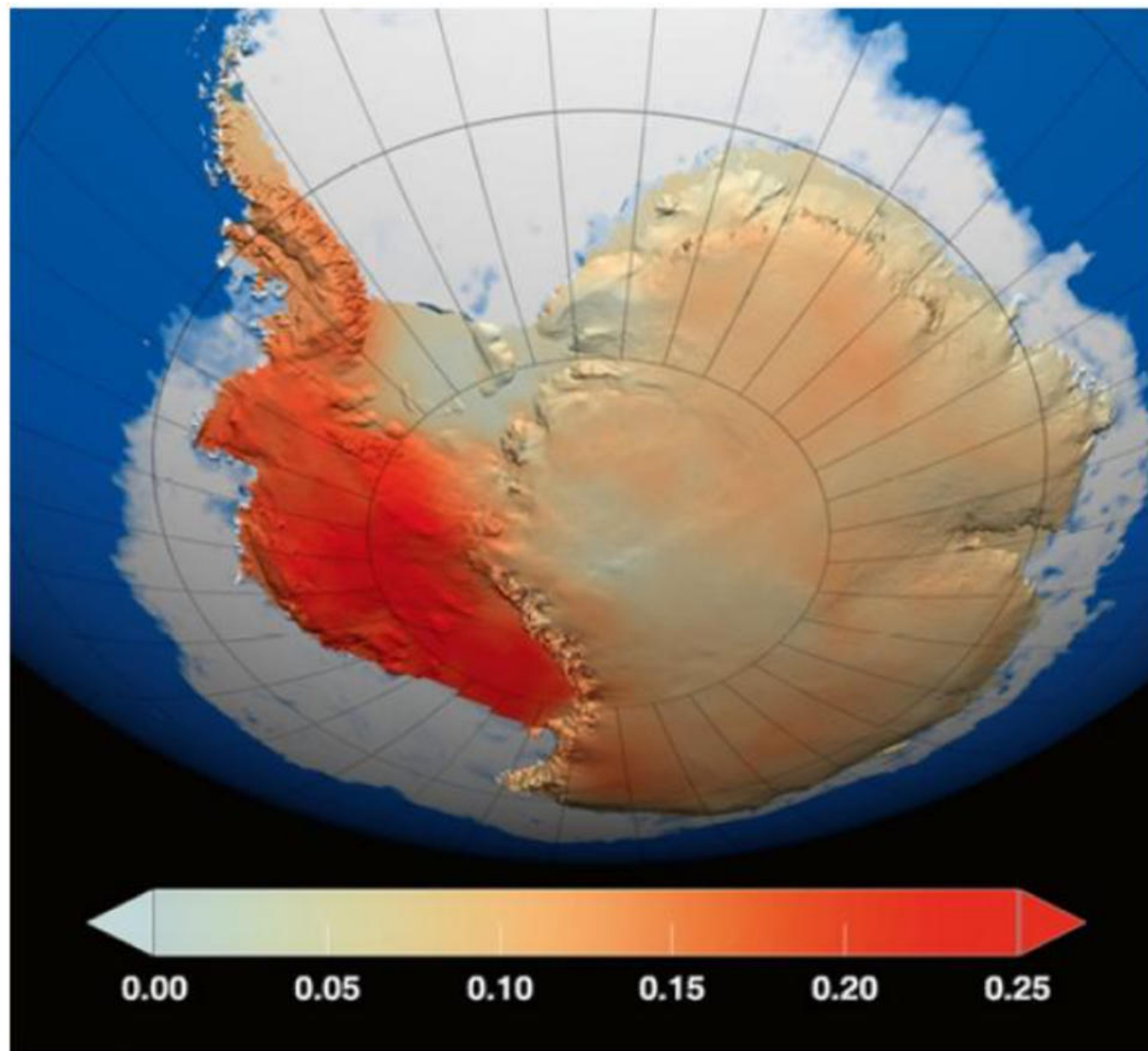
The principles and objectives of Article 9 of the 1959 Antarctic Treaty include:

- » the use of Antarctica for peaceful purposes only
- » the facilitation of scientific research in Antarctica

- » the facilitation of international scientific cooperation in Antarctica
- » the facilitation of the exercise of the right of inspection
- » questions relating to the exercise of jurisdiction in Antarctica
- » preservation and **conservation of living resources** in Antarctica.

The Treaty has been successful so far, partly due to the remoteness and isolation of Antarctica. Antarctica is the only continent on which there has not been a war. Its mineral resources have not been exploited, although its living natural resources such as whales, seals, fish and krill have been exploited. However, as Antarctica warms up and ice melts there could be increased pressure to develop its mineral resources.

Moreover, as global warming increases, the negative impacts on Antarctica are also likely to increase ([Figure 3.7](#)). Likewise, if the volume of tourists visiting the continent continues to increase, the impacts are likely to intensify. The same is true for pollution.



▲ **Figure 3.7** Antarctic warming since 1957 (°C)

Activities

- 1 Outline the main principles and objectives of the Antarctic Treaty.
- 2 Compare the potential impacts of low- and high-global warming emissions on Antarctica.

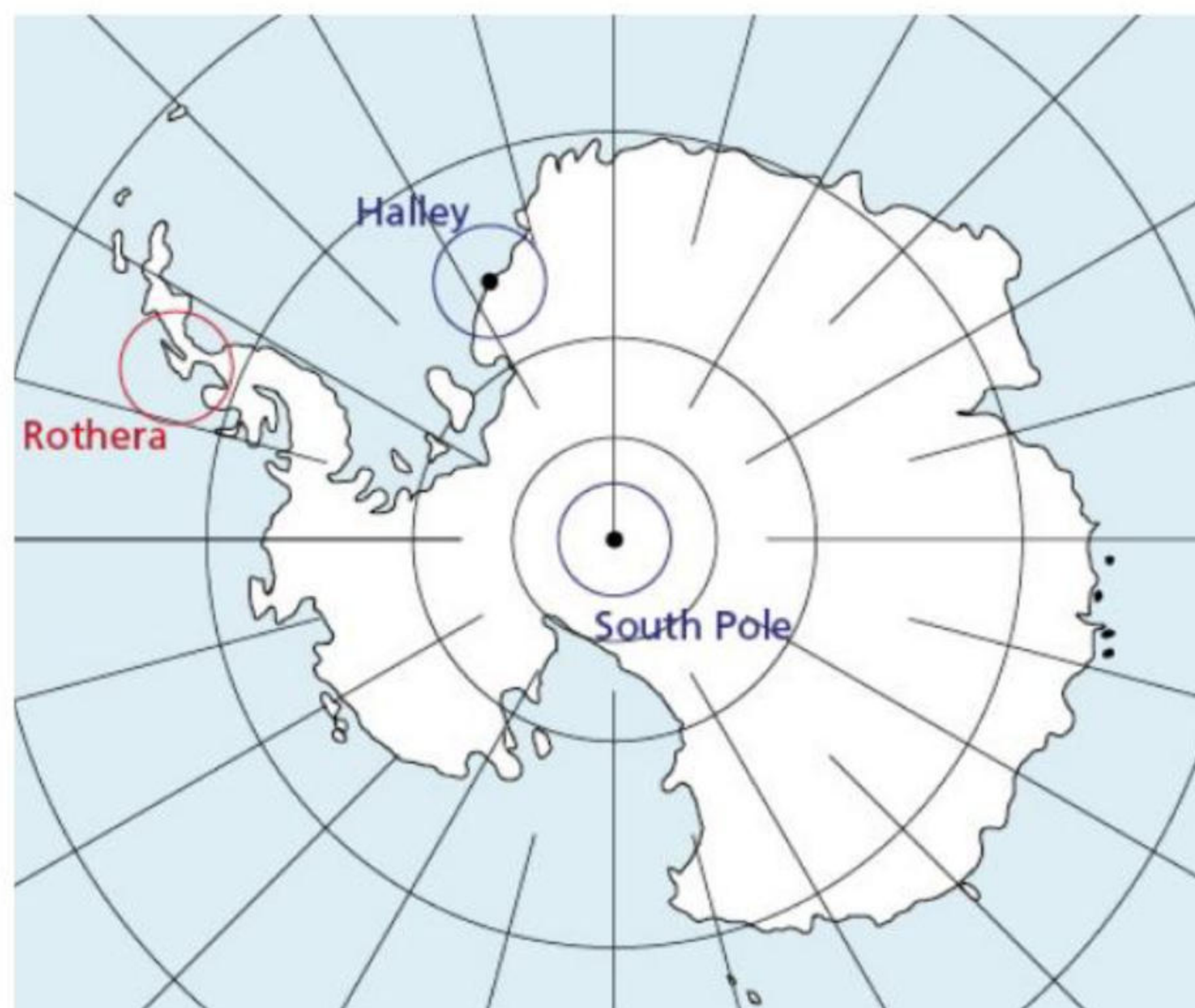
Environmental impact surveys

An **environmental impact survey** is a detailed survey to establish the likely impacts of a new development. Before any development starts, the environmental impact survey examines and identifies environments and species that are of high **conservation** importance or at risk from the development. These surveys make it possible to analyse the potential impacts of new developments on ecosystems and introduce management strategies to protect environments at risk.

By 2018, the wharf at the Rothera Research Station off the Antarctic Peninsula ([Figure 3.8](#)) was 25 years old and beyond repair. Moreover, more modern boats were longer and deeper than the wharf would allow to dock. The 'do nothing' and '**do minimum**' approaches were considered unsafe for the berthing of vessels, so a new wharf had to be constructed.



▲ **Figure 3.8** Rothera Research Station



▲ **Figure 3.9** The location of the Rothera Research Station, Antarctica

Some 24,000–30,000 m³ of local rock was sourced for the new wharf. Local material had to be used as the risk of introducing alien species was too high. Levels of biodiversity at Rothera were not high. Although seals were present, they did not breed in the area. The most significant environmental impacts were:

- » the introduction of non-native species
- » terrestrial and marine pollution from oil spills
- » the removal of rock
- » the removal of ice-free areas for terrestrial habitats
- » the disturbance of marine mammals by underwater noise.

The wharf was started in 2018 and completed in 2020.

Activities

- 1 Define the term 'environmental impact survey'.

2 Outline the potential impacts of new buildings on Antarctica’s environment.
